

The role of the dynamics of the norm. As discussed at the end of previous sub-section, after a rapid initial decrease of norm, the optimization resembles the dynamics of SGD projected onto a sphere. We stress that this is the crucial phase in training and the implicit regularization induced by SGD during this spherical optimization leads to better generalization. To validate this observation and isolate it from the initial norm drop, we investigate the behavior of SGD on a sphere with scale-invariant networks (Li & Arora, 2019). Scale invariance is chosen for its ease of LR tuning and for comparison with existing works (Kodryan et al., 2022; Li et al., 2020). Fig. 15 depicts a similar phenomenon as Fig. 3, where the test error vs. LR exhibits a U-shaped curve. While Kodryan et al. (2022) makes a similar observation, they do not provide an explanation. We demonstrate that the implicit regularization of the Jacobian norm is the key factor, elucidating its dependence on LR.

Effective learning rate vs. high training loss. Zhang et al. (2018); Van Laarhoven (2017) explored the relationship between LR and WD, introducing the concept of effective LR $\eta_e = \eta / \|\mathbf{w}\|_2^2$. These studies highlight that WD, preventing unbounded growth of the norm, enables the training process to evolve with a higher effective LR. This hypothesis is justified only with scale-invariance (which does not hold for general architecture). Furthermore, the underlying mechanism by which a higher LR enhances generalization is understood only in limited settings (Li et al., 2019). We propose that a high LR, combined with WD, leads to an increase in $\sigma_{\eta, \lambda}$. This hypothesis allows us to fully characterize and understand the mechanism through which generalization is enhanced.

Mixing in the function space. A simpler conjecture could have been stated in terms of the mixing of the iterates $(\mathbf{w}_t)_{t \geq 0}$ towards a solution of the regularized objective $\mathbf{w}_{\eta}^*$. However, Li et al. (2020) shows empirical evidence against mixing in the parameter space, emphasizing the necessity of considering the function space. Hence, our conjecture is formulated to capture stationarity in function space.

On the benefit of normalization. Our conjecture characterizes the mixing distribution but does not delve into the speed of the mixing process. In our experiments, we observe that normalization layers enable faster mixing. Li et al. (2020) observes a similar phenomenon in the case of scale-invariant networks, specifically the fast equilibrium conjecture, which is addressed by Li et al. (2022b). We note that this phenomenon persists even when the models are not exactly scale-invariant.

2.3 EMA and Fine-tuning

The large-LR phase sets the stage for SGD’s inherent biases to emerge but to actually exploit such bias, reducing the stochastic noise is necessary. This can be attained in two different ways: averaging (EMA) or decaying the learning rate (fine-tuning), both strategies are widely adopted in practice. This section illustrates the relation between the two and highlights the benefits of using one over the other and the implications for our analysis. From a practical standpoint, implementing EMA is more efficient than LR-decay because it does not require additional gradient iterations or hyperparameter tuning. While both methods enhance performance, their effectiveness is contingent on being combined with loss stabilization, supporting the hypothesis that the noisy dynamics is the underlying factor for their success. Although EMA shows only a slight advantage, our experiments in Fig. 5b, 2a, 2d demonstrate that it consistently outperforms learning rate decay in various settings.

When empirically validating Conjecture 1, the EMA is useful to characterize the limit point (i.e., $t \rightarrow \infty$) but cannot adequately capture the dynamics throughout the entire trajectory. This limitation arises because different points along the trajectory are at different loss values, making the comparison of any relevant regularized quantities inconsistent. An approach to overcome this inconsistency is to project the iterate $\mathbf{w}_t$ onto a manifold of constant loss. This can be achieved via early-stopped gradient flow (Li et al., 2021b) (SGD with small LR) on the CE loss with $\lambda = 0$ where $\mathbf{w}_t$ is projected to a nearby point $\tilde{\mathbf{w}}_t$, such that $\mathcal{L}(\tilde{\mathbf{w}}_t) \sim \text{const.}, \forall t$. In practice, this corresponds to fine-tuning via LR-decay. Since after fine-tuning $\mathcal{L}(\tilde{\mathbf{w}}_t) \approx \mathcal{L}(\tilde{\mathbf{w}}_{t'}), \forall t, t'$ see Fig. 5a, we can compare $\|J(\tilde{\mathbf{w}}_t)\|_F$ and $\|J(\tilde{\mathbf{w}}_{t'})\|_F$ and understand its evolution. In the experiments detailed in Fig. 5c, we report $\|J\|_F$ along the fine-tuned iterates $\tilde{\mathbf{w}}_t$ and observe a decreasing trend, i.e., the sequence $\{\|J(\tilde{\mathbf{w}}_t)\|_F\}_{t \geq 0}$ is decreasing. This fact empirically validates that the entire trajectory of the iterates $(\mathbf{w}_t)_{t \geq 0}$, closely following the trajectory of the fine-tuned iterates $(\tilde{\mathbf{w}}_t)_{t \geq 0}$, bias the model towards a regularized solution that might enhance generalization. This also explains why learning rate schedules, such as step-decay, which starts with a large value and then decrease it, can enhance generalization.

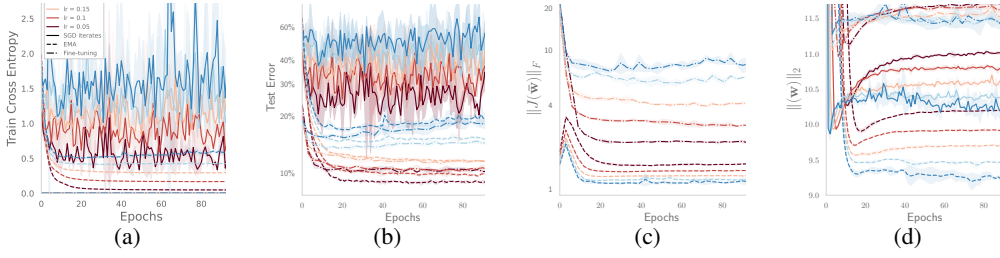


Figure 5: **EMA vs Fine-tuning.** Training of standard Resnet18 on CIFAR-10 for 100 epochs fixing $\lambda = 0.0125$ and varying the learning rate. In Fig. 5a we report different levels of loss stabilization, in Fig. 5b we report the test errors and in Fig. 5c and Fig. 5d the norm of the Jacobian and of the weights respectively. The quantities are measured for the SGD iterates, the EMA and the fine-tuning. The latter is performed for 100 epochs every 3 with $\eta = 10^{-3}$.

Despite providing a straightforward methodology to analyze the trajectory, LR-decay introduces additional complexities that cause deviations from the conjecture. Indeed, in Fig. 5c we observe that the final points of the fine-tuned iterates report the opposite trend compared to the EMA (smaller η lead to larger $\|J\|_F$). This discrepancy is potentially due to the state-dependent nature of the SGD noise covariance in equation 3; decreasing the LR and removing WD can alter the stationary distribution and the regularized objective, leading to a different solution than anticipated by Conjecture 1.

3 Weight decay in the under-training regime

In this section, we investigate how WD enhances optimization in the under-training regime. Although the phenomenon is more general, we focus on LLMs for which one-epoch training is typically used.

Two key effects of weight decay in the under-training regime. WD is widely used in training state-of-the-art LLMs like GPT-3, Chinchilla, and Llama (Brown et al., 2020; Hoffmann et al., 2022; Touvron et al., 2023). While Brown et al. (2020) suggest that WD offers “a small amount of regularization,” its necessity remains unclear in the context of *one-pass* training where the population loss is directly minimized. As a sanity check, in Fig. 19 in Appendix, we verify that the generalization gap is close to zero even for models trained without WD. Instead of the regularization effect, we suggest that the two most crucial effects of WD in the under-training regime are (1) better optimization of the training loss as briefly observed by Hoffmann et al. (2022), (2) prevention of loss divergences under the `bf16` weight precision. We reproduce this phenomenon at a smaller scale with 124M parameters in Fig. 6: the final training loss is smaller for λ equal to 0.1 and 0.3 compared to 0. We study both mechanisms which stand in contrast to the over-training regime of Sec. 2, where the primary concerns are not optimization and stability, but rather generalization.

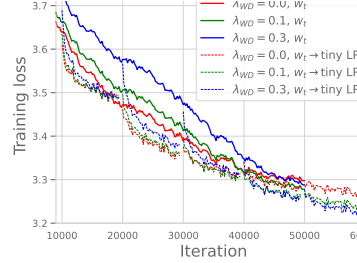


Figure 6: **GPT-2-124M on OpenWebText.** We reproduce the improvement from WD as in Hoffmann et al. (2022) but at a much smaller scale. Performing fine-tuning with a tiny LR reveals that a higher starting training loss can still be a better point in terms of the final loss after fine-tuning.

Experimental setup. We use the NanoGPT repository (Karpathy, 2023) for training GPT-2 models (Radford et al., 2019) on OpenWebText. We train a 124M parameter model (known as GPT-2-Small) for 50 000 iterations with a batch size of 256. For most experiments, we reduce the default context length from 1024 to 256 to ensure practicality within an academic budget. Alternatively, we could have reduced the number of training iterations or batch size, but this would lead to insufficiently trained models. Unless mentioned otherwise, we train with AdamW using the default LR 0.0006, a short 400-iteration LR warmup, gradient clipping with the ℓ_2 -threshold 1.0, and $10\times$ cosine LR decay. We keep all other hyperparameters at their default values (see App. B).

Better optimization with WD is reproducible at a smaller scale. The findings from Hoffmann et al. (2022) (Fig. A7 therein) indicate that WD in AdamW leads to lower training loss (≈ 0.02 lower), primarily towards the end of training. The reduction of training loss directly translates to a better downstream performance and makes this observation practically relevant. Additionally, performing fine-tuning with a tiny LR reveals that a higher starting training loss can still be a better starting point

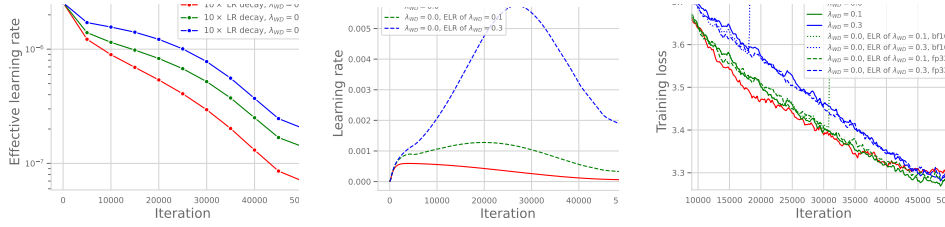


Figure 7: **GPT-2-124M on OpenWebText.** *Left:* The effective LR $\eta_t / \|\mathbf{w}_t\|_2$ for the models reported in Fig. 6. *Middle:* The LR schedule that matches the effective LR $\eta_t / \|\mathbf{w}_t\|_2$ of the runs with weight decay 0.1 and 0.3. *Right:* Matching the effective LR is sufficient to match the whole training dynamics of the loss if we avoid the loss spikes by using full precision (float32 instead of bfloat16).

in terms of the final loss after fine-tuning. Moreover, in Fig. 21, we show that decoupling WD, as advocated by Loshchilov & Hutter (2019), is not necessary to achieve this effect: a simple ℓ_2 penalty added to the loss suffices. Lastly, in Fig. 22, we show that a similar improvement in training loss is observed for *SGD with momentum* suggesting that adaptive LR are not key for this phenomenon.

Effective LR induced by weight decay in AdamW. We hypothesize that the use of WD for LLM training results in an increased effective LR, even in the absence of scale invariance of the training loss for modern transformer architectures. Here we show that WD in combination with sign SGD—utilized as a surrogate for Adam (Balles & Hennig, 2018)—is equivalent to projected SGD on the sphere, with an effective LR $\eta_{\text{eff}} \propto \eta_t / \|\mathbf{w}_t\|_2$, similarly to Van Laarhoven (2017). Consider the update rule of sign SGD on loss ℓ with WD:

$$\mathbf{w}_{t+1} = (1 - \eta_t \lambda_t) \mathbf{w}_t - \eta_t \text{sign}(\nabla \ell_t(\mathbf{w}_t)) = (1 - \eta_t \lambda_t) \|\mathbf{w}_t\|_2 \left[\frac{\mathbf{w}_t}{\|\mathbf{w}_t\|_2} - \frac{\eta_t \cdot \text{sign}(\nabla \ell_t(\mathbf{w}_t))}{(1 - \eta_t \lambda_t) \|\mathbf{w}_t\|_2} \right].$$

Considering the evolution of the direction $\tilde{\mathbf{w}} := \mathbf{w} / \|\mathbf{w}\|_2$,

$$\tilde{\mathbf{w}}_{t+1} \propto \left[\tilde{\mathbf{w}}_t - \frac{\eta_t}{(1 - \eta_t \lambda_t) \|\mathbf{w}_t\|_2} \cdot \text{sign}(\nabla \ell_t(\mathbf{w}_t)) \right].$$

When $\text{sign}(\nabla \ell_t(\mathbf{w}_t))$ is determined solely by the direction $\tilde{\mathbf{w}}_t$, the evolution of the direction of weights becomes the primary matter. This scenario occurs when the function ℓ is scale-invariant or homogeneous. Observing the trend of the gradient norm and the parameter norm $\|\mathbf{w}_t\|_2$ from Fig. 26, we note an inverse relationship, i.e., the gradient norm is higher when the parameter norm is lower. This behavior is reminiscent of scale-invariant networks for which $\nabla \ell(\alpha \mathbf{w}) = \frac{1}{\alpha} \nabla \ell(\mathbf{w})$, for any $\alpha \neq 0$. Thus, controlling parameter norms with WD allows implicit changes to the LR schedule which we verify experimentally next.

Matching the effective LR without weight decay. To verify the key role of the ELR $\eta_t / \|\mathbf{w}_t\|_2$, we train a model without WD but with ELR corresponding to models trained with $\lambda \in \{0.1, 0.3\}$ and report results in Fig. 7. We observe that the *whole* training dynamics is matched which confirms our hypothesis. This fully explains the observation from Hoffmann et al. (2022) about the advantage of AdamW over Adam: there exists an LR schedule (albeit a non-standard one) shown in Fig. 7 (middle) that leads to the same loss profile as the original AdamW run. However, we note that this holds only for full float32 precision, and models trained with bfloat16 precision diverge in the middle of training. This experience suggests that WD is still necessary in practice to prevent loss divergence. We also note that matching the ELR $\eta_t / \|\mathbf{w}_t\|_2$ derived above for sign SGD instead of $\eta_t / \|\mathbf{w}_t\|_2^2$ for plain SGD (Zhang et al., 2018; Hoffer et al., 2018) is critical for AdamW. Otherwise, the runs diverge very early in training, even with float32 parameter precision.

Explaining the training dynamics of AdamW. Classical optimization theory suggests that convergence of SGD-based algorithms primarily depends on two factors: the *bias* term that influences the rate at which initial conditions are forgotten and the *variance* term that results from noise in the gradient estimates (Moulines & Bach, 2011). We argue that these two factors, together with the observation about higher ELR for WD, can explain the loss profiles from Fig. 6. If we consider the simple case of SGD with a constant LR η applied to a linear least-squares problem, the expected excess risk after t iterations can be bounded as a sum of a bias and variance terms:

$$\text{Excess Risk} \lesssim (1 - \eta \mu)^t \|\mathbf{w}_0 - \mathbf{w}_*\|^2 + \eta \sigma^2,$$